Name	NRIC Number	CTG

YISHUN JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION 2016

BIOLOGY **9648/01**

HIGHER 2

29 AUGUST 2016
MONDAY 0800 – 0915

Paper 1 Multiple Choice **1 hour 15 minutes**

Additional material:
Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

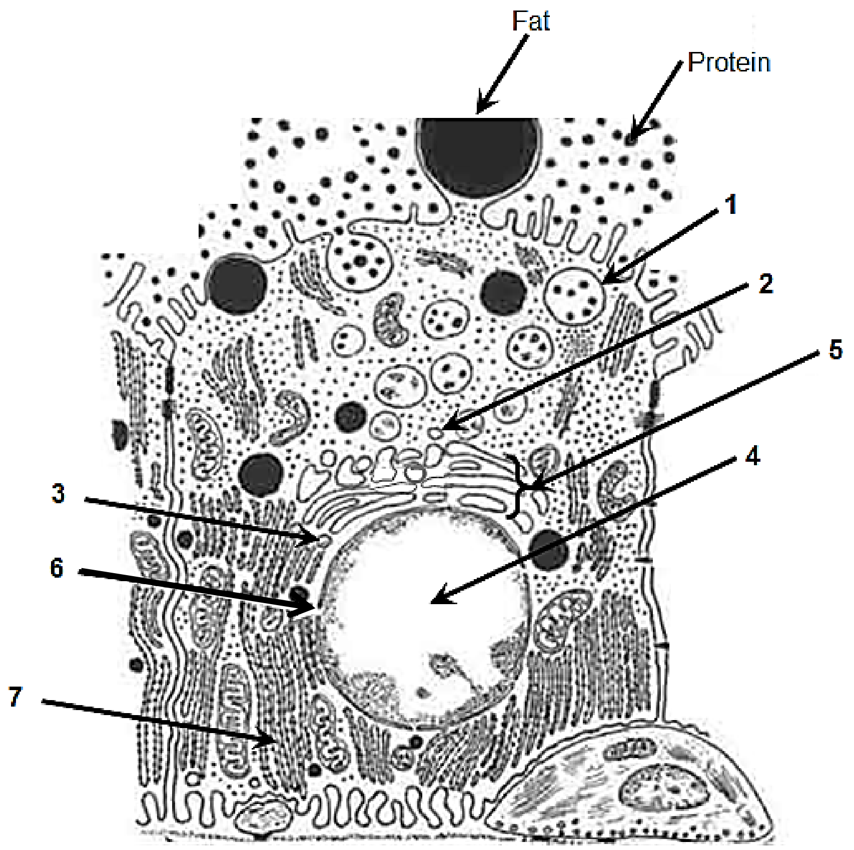
Write in soft pencil (type B or HB is recommended).
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, full NRIC number and CTG on the Multiple Choice Answer Sheet and question paper in the spaces provided.

There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Optical Mark Sheet.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer. Any rough working should be done in this booklet.

Calculators may be used.

1 The diagram below shows a lactating mammary secretory cell. In this cell, milk proteins, such as casein, are transcribed and translated, and eventually secreted into the lumen of the mammary gland.



Which of the following shows the most likely sequence of locations involved in this process?

	start			→			finish
A	6	3	4	7	2	5	1
B	6	4	3	7	5	2	1
C	4	6	7	3	5	2	1
D	4	3	7	6	2	5	1

2 The table gives description of four membranous structures in a cell.

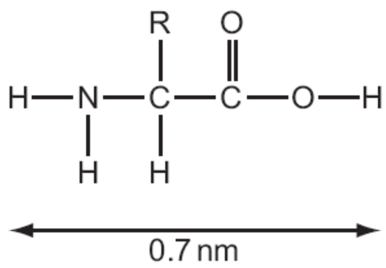
Which structure is correctly matched to its function?

	Structure	**Function**
A	an extensive network of tubes and sacs; each tube and sac bounded by a single membrane	Lipid synthesis
B	a spherical sac bounded by a single membrane	Protein synthesis
C	a sac bounded by two membranes, the inner highly folded	Packaging of proteins
D	a stack of elongated, curved sacs; each sac bounded by a single membrane	Photosynthesis

3 The diameters of some atoms when they form bonds are given in the table.

Atom	Single bond / nm	Double bond / nm
H	0.060	-
O	0.132	0.110
N	0.140	0.120
C	0.154	0.134

The approximate length of the amino acid shown below was estimated using the figures in the table.



What will be the approximate length of a dipeptide formed using this amino acid?

A 0.8 nm **B** 1.2 nm **C** 1.5 nm **D** 1.9 nm

4 During the production of fruit juice, enzymes are used to break down the components of cell walls. Which carbohydrate will be produced by this hydrolysis?

A Sucrose **B** Maltose **C** α – glucose **D** β – glucose

5 The enzyme phosphofructokinase is involved in phosphorylation of hexose phosphate sugars during glycolysis. It is involved in control of the rate of glycolysis and thus respiration, by end-product inhibition.

Deduce which of the following is a description of this enzyme.

	Shape of binding site(s)	Substrate	Product
A	no allosteric site, active site complementary to ATP and hexose	hexose	hexose phosphate
B	allosteric site complementary to glucose, active site complementary to hexose phosphate	hexose phosphate	hexose phosphate
C	allosteric site complementary to ATP, active site complementary to ATP and hexose phosphate	hexose phosphate	hexose bisphosphate
D	no allosteric site, active site complementary to hexose bisphosphate	hexose bisphosphate	two triose phosphate

6 A cell in the G1 phase has two homologous pairs of chromosomes. It then undergoes a mitotic division, followed by meiosis. At the end of meiosis II, what is the total number of chromosomes and gene loci found in all the daughter cells formed?

A 8 chromosomes and 4 times as many gene loci as the original parent cell

B 8 chromosomes and 8 times as many gene loci as the original parent cell

C 16 chromosomes and 4 times as many gene loci as the original parent cell

D 16 chromosomes and 8 times as many gene loci as the original parent cell

7 For organisms undergoing sexual reproduction, a reduction division occurs before fertilisation.

Which reason(s) explain why this is necessary?

1 increase genetic variation
2 prevent doubling of the chromosome number
3 reduce the chances of mutation

A 1 only **B** 2 only **C** 1 and 2 only **D** 1, 2 and 3

8 One complete turn of the double helix of DNA contains 10 pairs of bases and is 3.4nm long. What is the approximate number of amino acids in an enzyme coded by a 132 nm length of DNA?

A 38

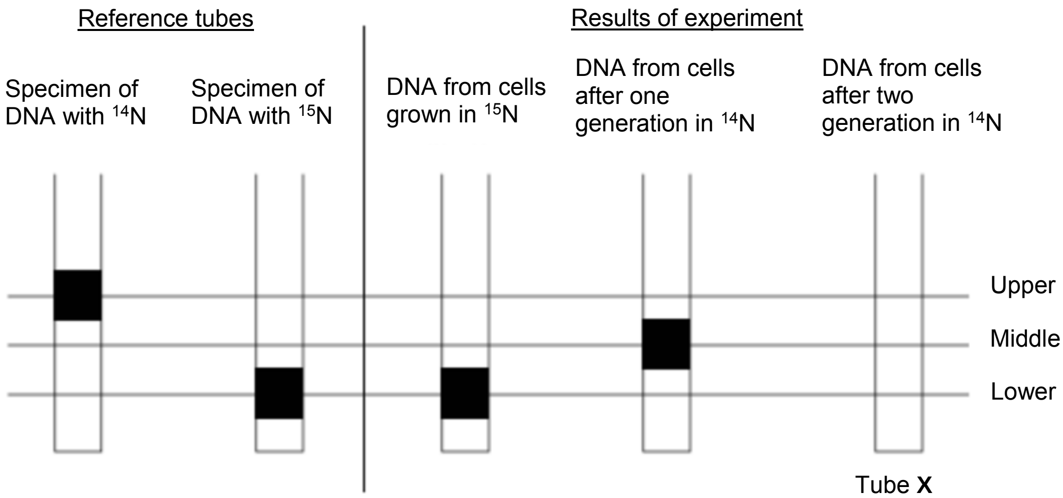
B 129

C 150

D 388

9 Cells of the bacterium E. coli were grown for many generations on a medium containing only the heavy isotope of nitrogen, ^{15}N. The cells were then transferred to a medium containing only ^{14}N and allowed to grow. Samples of the bacteria were removed from the culture after one generation and after two generations. The DNA from each sample was extracted and centrifuged.

The figure below shows two reference tubes and the results of this experiment.



What are the positions and relative proportions of the bands in Tube **X**?

	Upper	**Middle**	**Lower**
A	50%	50%	0%
B	0%	50%	50%
C	50%	0%	50%
D	25%	50%	25%

10 A student obtained a sample of DNA and an mRNA was transcribed from this DNA. The samples were subsequently purified. He then separated the two strands of the DNA sample.

The base compositions of each strand and that of the mRNA were analysed. The results of the analysis are shown in the table below.

	A	G	C	T	U
DNA strand 1	19.1	26.0	31.0	23.9	0.0
DNA strand 2	24.2	30.8	25.7	19.3	0.0
DNA strand 3	20.5	25.2	29.8	24.5	0.0
mRNA	19.0	25.9	30.8	0.0	24.3

A Strand **1**

B Strand **2**

C Strand **3**

D None of the above

11 Which of the following occurs in the life cycle of an influenza virus but **not** in the life cycle of HIV?

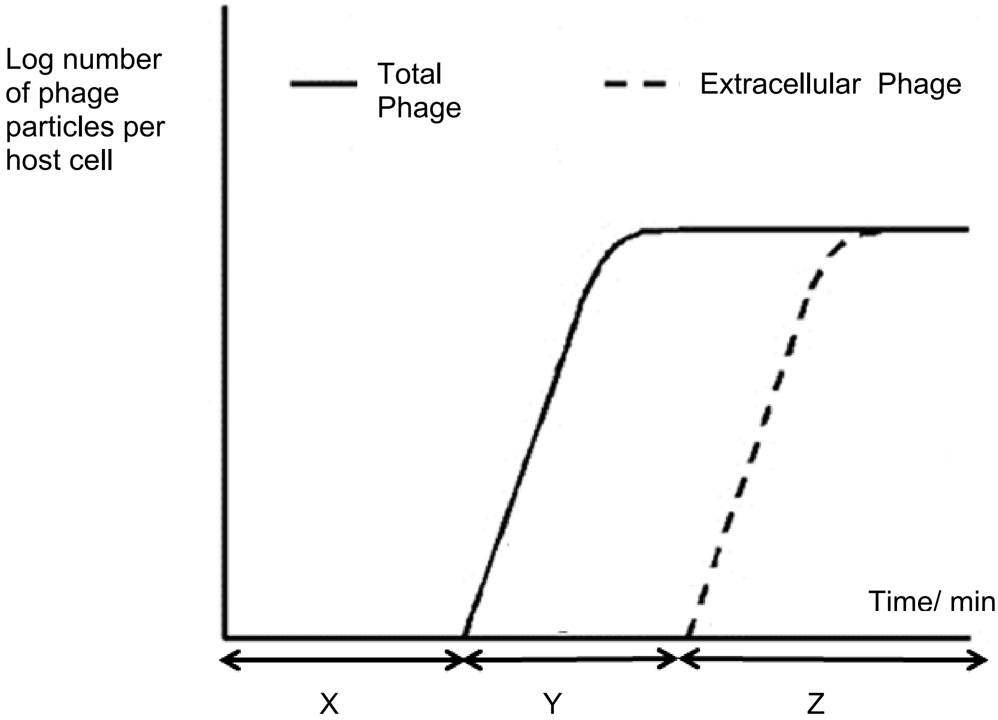
A Synthesis of double stranded complementary DNA.

B Integration of DNA into host genome.

C Binding to specific surface receptors followed by fusion of membrane.

D Removal of specific surface receptors during release.

12 The figure below shows a growth cycle of a T4 phage.



Which of the following statements about X, Y and Z of the growth cycle is correct?

A X is the eclipse period where the phage's tail fibers attach to specific receptors on the bacterial cell.

B Period Z will correspond to the death of host cells.

C X corresponds to the period where the phage exists as a prophage

D Y is the period where there is just active viral DNA replication and protein production.

13 Which statement correctly describes the control of transcription of the genes involved in the synthesis of tryptophan in Escherichia coli?

A A repressor protein is inactive and does not bind to the operator, switching on the genes.

B A repressor protein is active and does not bind to the operator, switching off the genes.

C A repressor protein is active and binds to the promoter, switching on the genes.

D A repressor protein is inactive and does not bind to the promoter, switching off the genes.

14 A similarity between transformation and conjugation in bacteria is that they

A both involve double-stranded DNA passing into the cell whereby one strand is degraded by nucleases.

B both involves homologous recombination where nucleotide sequences are exchanged.

C both involves the formation of sex pili in order for the DNA to enter.

D both involve recombination and forming new plasmids.

15 Some events that take place during generalised transduction are listed below.

1 Bacterial host DNA is fragmented
2 Bacterial DNA may be packaged in a phage capsid
3 Recombination between donor bacterial DNA and recipient bacterial DNA
4 Phage infects a bacterial cell
5 Phage DNA and proteins are made

Which sequence of events is correct?

A 4 1 5 2 3

B 4 1 3 5 2

C 4 3 1 5 2

D 4 5 1 3 2

16 The following are all levels of control of gene expression.

1 Transcriptional
2 Post-transcriptional
3 Translational
4 Post-translational

Which sequence of events is correct?

A 1, 2, 3 and 4

B 1 and 2 only

C 2 and 4 only

D 3 and 4 only

17 Which statement correctly describes a role of histone proteins?

A All eukaryotic genes are transcribed continuously because they are not packaged by histones.

B DNA must be selectively released from its histone packaging before transcription can occur in bacteria.

C Histones package prokaryote chromatin into the nucleosomes that form the bulk of the chromosome.

D The organisation of DNA by histones in eukaryotes allows some gene control sequences to be thousands of base pairs away from the gene concerned.

18 Four different genes are regulated in different ways.

1 Gene 1 undergoes tissue-specific patterns of alternative splicing

2 Gene 2 is part of a group of structural genes controlled by the same regulatory sequences

3 Gene 3 is in some circumstances subject to methylation

4 Gene 4 codes for a repressor protein which acts at an operator site close by

Which combination correctly identifies the gene regulatory steps involving prokaryotes and eukaryotes?

	prokaryotic	eukaryotic
A	1 and 2	3 and 4
B	1 and 3	2 and 4
C	2 and 3	1 and 4
D	2 and 4	1 and 3

19 Ras protein is a G-protein involved in a cell-signalling pathway. This protein is coded for by a proto-oncogene. Which of the following would lead to the cell undergoing uncontrolled cell division?

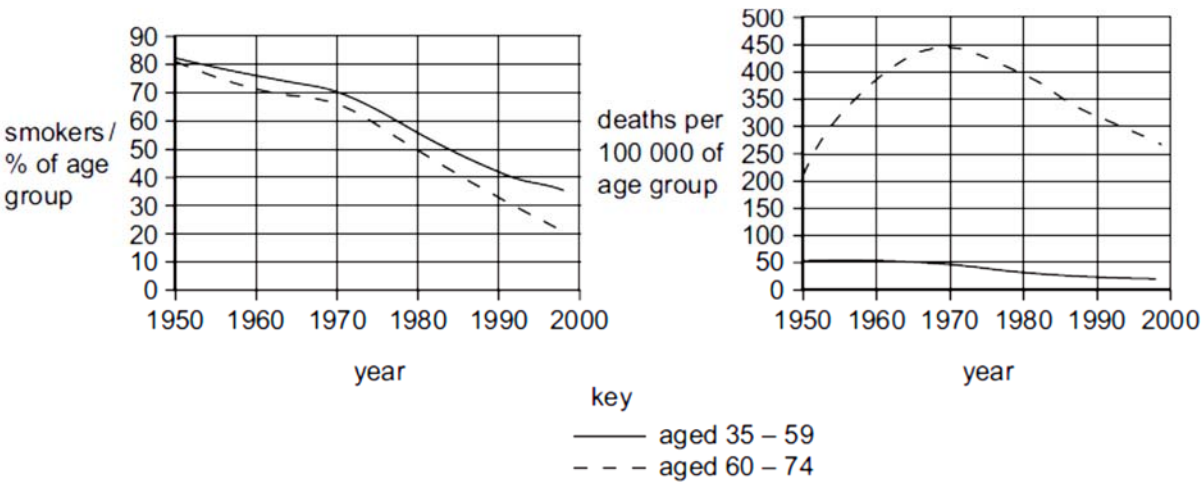
A Point mutation within the proto-oncogene, forming mutant Ras which is constitutively active.

B Formation of multiple copies of the Ras proto-oncogene resulting in mutant Ras proteins.

C Duplication of the Ras promoter that controls the proto-oncogene.

D Translocation of the proto-oncogene upstream of a hyperactive promoter.

20 Some studies suggest that smoking increases the risk of developing lung cancer. The two graphs show the percentage of smokers and the deaths from lung cancer in men of two age groups between 1950 and 1998.



Which statement is **not** supported by the data in the graphs?

A Deaths from lung cancer in men 35-59 decreased by 50 % over the period of the study.

B Deaths from lung cancer in men 60-74 increased up to 1970.

C The data for men 60-74 between 1950 to 1970 suggests that lung cancer takes up to 20 years to develop.

D The number of men aged 35-59 who were smokers decreased by approximately 70 % over the period of the study.

21 A parent organism of unknown genotype is mated in a test cross. Half of the offspring have the same phenotype as the parent.

What can be concluded from this result?

A The parent is heterozygous for the trait.

B The parent is homozygous dominant for the trait.

C The parent is homozygous recessive for the trait.

D The trait being inherited is polygenic.

22 The table shows the results of a series of crosses in a species of small mammal.

coat colour phenotype		
male parent	female parent	offspring
dark grey	light grey	dark grey, light grey, albino
light grey	albino	light grey, white with black patches
dark grey	white with black patches	dark grey, light grey
light grey	dark grey	dark grey, light grey, white with black patches

What explains the inheritance of the range of phenotypes shown by these crosses?

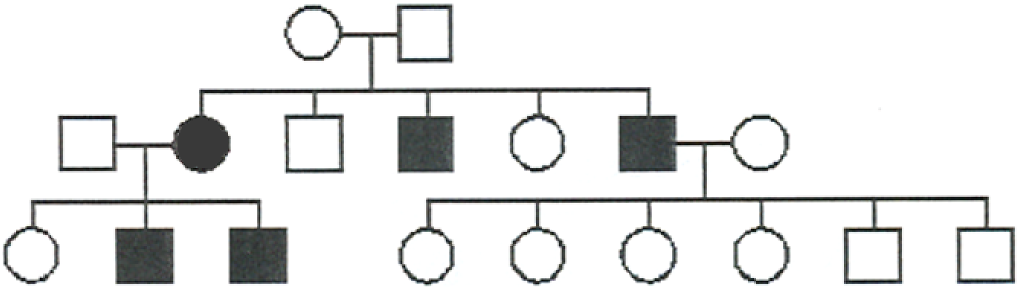
A one gene with a pair of co-dominant alleles

B one gene with multiple alleles

C sex linkage of the allele for grey coat colour

D two genes, each with a dominant and recessive allele

23 The following pedigree depicts the inheritance of a rare hereditary disease affecting muscles.



What is the mode of inheritance of this disease?

A autosomal dominant

B autosomal recessive

C X-linked dominant

D X-linked recessive

24 The statements are descriptions of aspects of genetics.

1 The phenotype is affected by both alleles at the same locus of a heterozygous individual.

2 The combined effects of alleles at two or more gene loci equal the sum of their individual effects.

3 Many different alleles present in a gene pool can occupy the same gene locus.

4 Alleles of one gene mask the effects of the alleles of another gene at a different locus.

Which of the following correctly describes the statements?

	Codominant alleles	Epistatic alleles	Multiple alleles	Additive genes
A	1	4	2	3
B	1	4	3	2
C	2	1	4	3
D	3	1	4	2

25 In cats, the genes controlling coat-colour are co-dominant and carried on the X chromosomes. When a black female was mated with a ginger male the resulting litter consisted of black male and tortoise-shell female kittens.

What phenotypic ratio would be expected in the F2 generation?

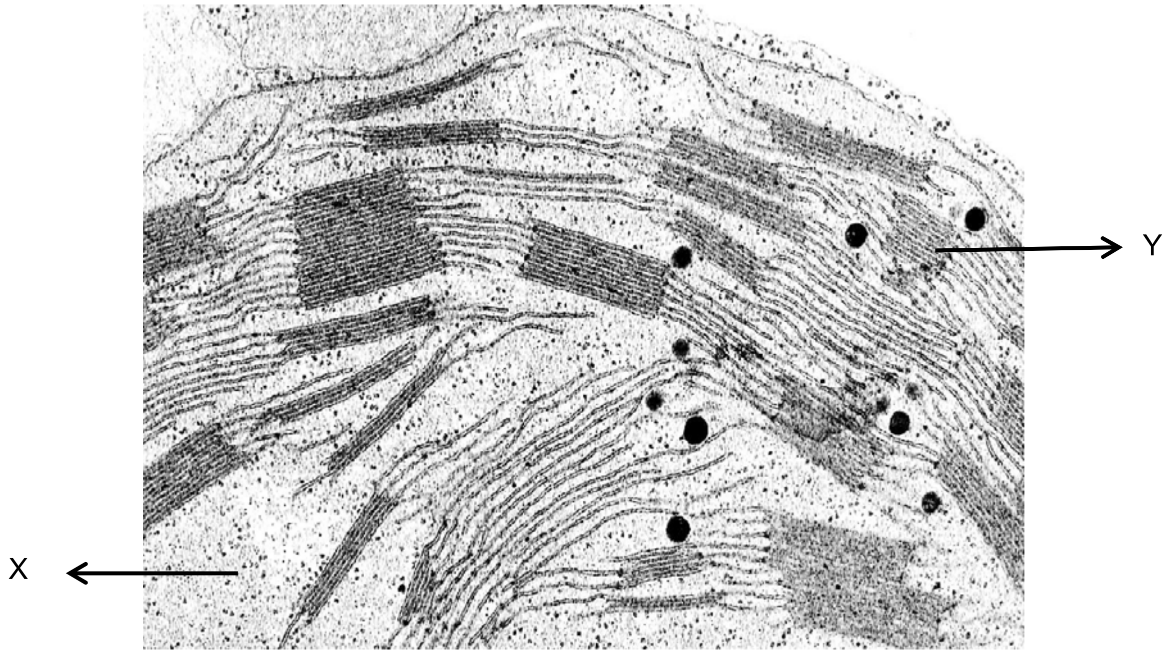
A 1 black male : 1 ginger male : 2 tortoise-shell females

B 1 black male : 1 ginger male : 1 tortoise-shell female : 1 black female

C 2 black males : 1 tortoise-shell female : 1 ginger female

D 2 black males : 1 tortoise-shell female : 1 black female

26 The electron micrograph shows a chloroplast of a plant cell with structures X and Y labelled.



Which of the following statements is correct?

A There is production of NADH and ATP within structure Y.

B There are photosynthetic pigments present in structure X which harnesses chemical energy

C Structure Y is the location in which triose phosphate is produced.

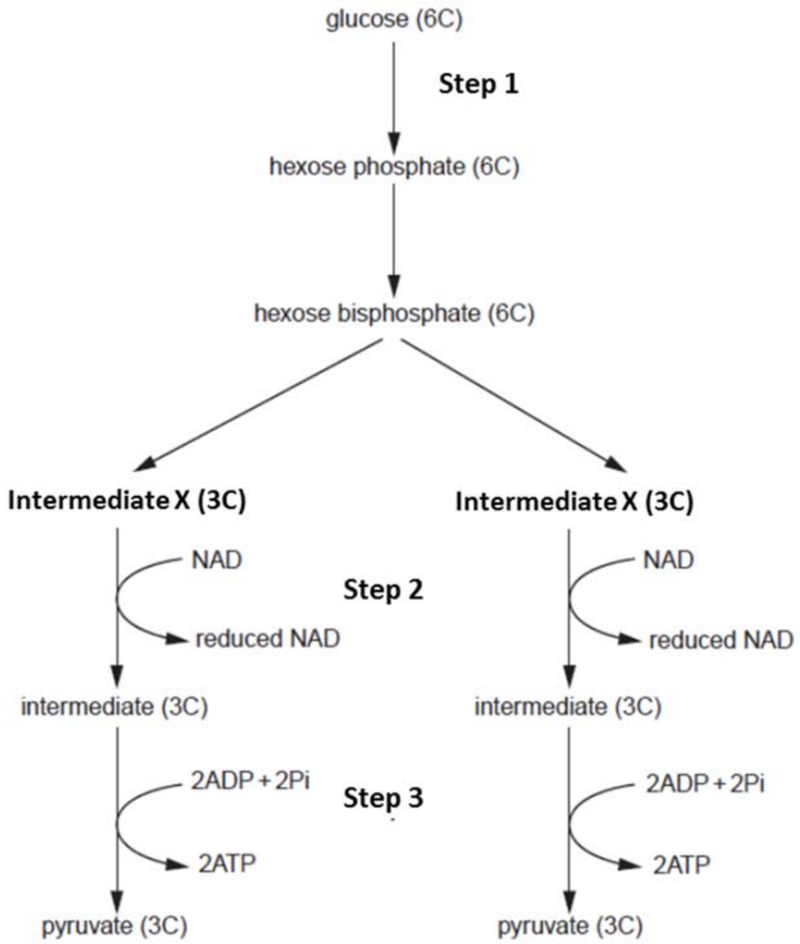
D Structure Y contains proteins involved in the phosphorylation of ADP.

27 2,4-dinitrophenol (DNP) is a chemical and was used for weight loss. However, the prolonged overdose of DNP leads to toxicity including the potential for hyperthermia and death. DNP targets the inner mitochondrial membrane to uncouple oxidative phosphorylation from electron transport. It allows protons to cross the inner mitochondrial membrane and thus dissipates the proton gradient. It also increases tissue metabolism.

Which of the following is true of a cell treated with 2,4-dinitrophenol?

	Ability to use oxygen	**Ability to produce carbon dioxide**	**ATP yield**
A	Yes	No	Decreases
B	Yes	Yes	Decreases
C	No	Yes	Increases
D	Yes	No	Increases

28 During glycolysis, glucose is converted by a series of steps into two molecules of pyruvate.



Which of the following correctly labels intermediate X, steps 1 – 3 and the location where step 3 occurs at?

	Location	**Intermediate X**	**Step 1**	**Step 2**	**Step 3**
A	Cytosol	Glyceraldehyde-3-phosphate	Reduction	Oxidation	Oxidative phosphorylation
B	Cytosol	Triose phosphate	Phosphorylation	Oxidation	Substrate level phosphorylation
C	Mitochondrial matrix	Glycerate-3-phosphate	Reduction	Reduction	Substrate level phosphorylation
D	Cytosol	Triose phosphate	Phosphorylation	Reduction	Substrate level phosphorylation

29 A scientist is investigating the effects of Poison T on the cell signalling pathway of glucagon. It is found that Poison T diminishes the effect of glucagon.

The table below shows the different components of the cell signalling pathway of glucagon and their statuses.

Component	**Status**
G Protein Coupled Receptor (GPCR)	Configuration can be changed
G protein	Can exchange GDP for GTP
cAMP levels	low
Protein Kinase A	Inactivated

Which of the following statements can be concluded from the results?

1 Poison T interferes with reception.

2 Poison T prevents G protein from hydrolysing GTP.

3 Poison T inactivates the enzyme adenylate cyclase.

4 Poison T stops the relaying of message down the phosphorylation.

A 1 and 2

B 2 and 3

C 2 and 4

D 3 and 4

30 Impulses travel very rapidly along nerves from the leg muscles of a mammal because

A there is a high concentration of Na+ ions inside the axons.

B the nerves contain myelinated fibres.

C there is a potential difference across the axon membranes

D a nerve impulse is an all-or-nothing phenomenon.

31 As adults, certain species of whales possess baleen instead of teeth. Baleen is used to filter the whales' diet of planktonic animals from seawater. As embryos, baleen whales possess teeth, which are later replaced by baleen. The teeth of embryonic baleen whales are evidence that

A All whales are descendants of terrestrial mammals.

B Baleen whales are descendants of toothed whales.

C Baleen embryos pass through a stage when they resemble adult toothed whales.

D Among ancient whales, baleen evolved before teeth.

32 Which of the following increases the number of different alleles in a population?

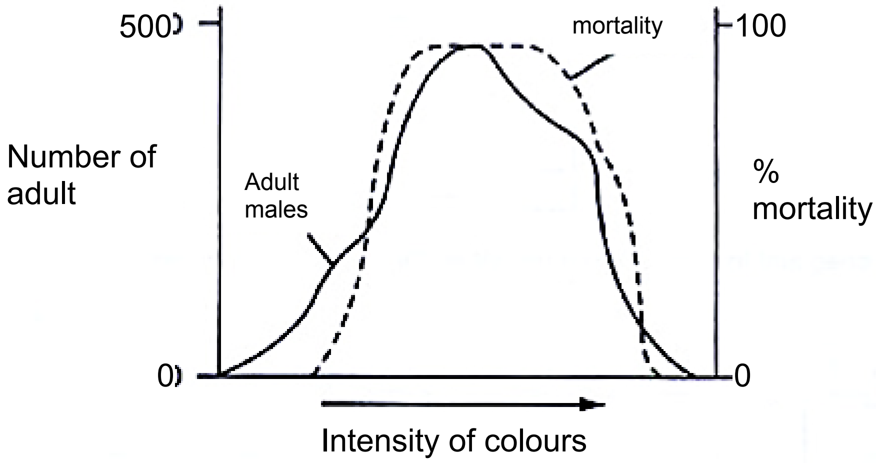
A crossing over

B gene mutation

C random fusion of gametes

D random assortment of chromosomes in meiosis

33 The graph below shows data on a population of a species of moth which shows considerable variation in colour intensity. Which conclusion can be made from this graph?



A Colour variation is environmentally induced.

B Colour variation is genetically determined.

C Extreme forms are favoured by natural selection.

D The species shows discontinuous variation with respect to colour.

34 Which uses of the information from the human genome project are generally considered to be unethical?

1 an insurance company only giving cheap rates to people with genetic predispositions to fewer diseases

2 genetic archaeologists identifying the earliest forms of genes to show evolutionary relationships

3 cytologists developing tests for only some defective genes

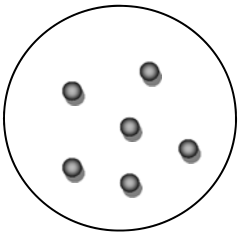
4 doctors only giving specific drugs to block the actions of faulty genes to carriers of those genes

5 genetic counsellors giving specific lifestyle information only to people genetically predisposed to risks

6 parents choosing embryos for implantation only after ante-natal tests for acceptable genes

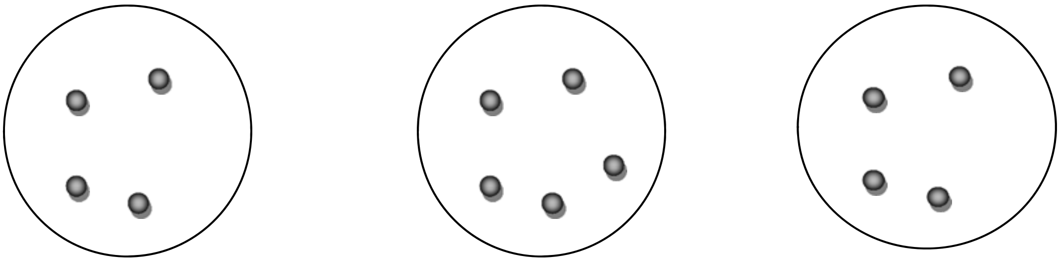
A 1 and 3 **B** 1 and 6 **C** 2 and 5 **D** 3 and 4

35 The insulin gene and a plasmid with ampicillin and tetracycline resistant genes were cut using the same restriction enzyme, EcoRI. The cut plasmid and insulin gene was mixed together and DNA ligase was allowed to react. Subsequently the mixture was heat shocked at 42 degrees with E. coli. The mixture was then plated on nutrient agar to obtain a master plate with colonies shown below.



Master plate (nutrient agar)

Three replica plates containing different antibiotics were obtained from the master plate. The diagram below shows the results of the replica plates.



Nutrient agar with ampicillin | **Nutrient agar with tetracycline** | **Nutrient agar with tetracycline and ampicillin**

Which of the following can be concluded from the results obtained?

A The ampicillin resistant gene is inactivated by insertion of the insulin gene

B The tetracycline resistant gene is inactivated by insertion of the insulin gene.

C The tetracycline resistant gene does not act as a selectable marker

D The ampicillin resistant gene does not act as a selectable marker.

36 Which of the following is the most powerful way of increasing the specificity of a DNA profile analysis?

A Repeat the analysis multiple times.

B Increase the number of markers used.

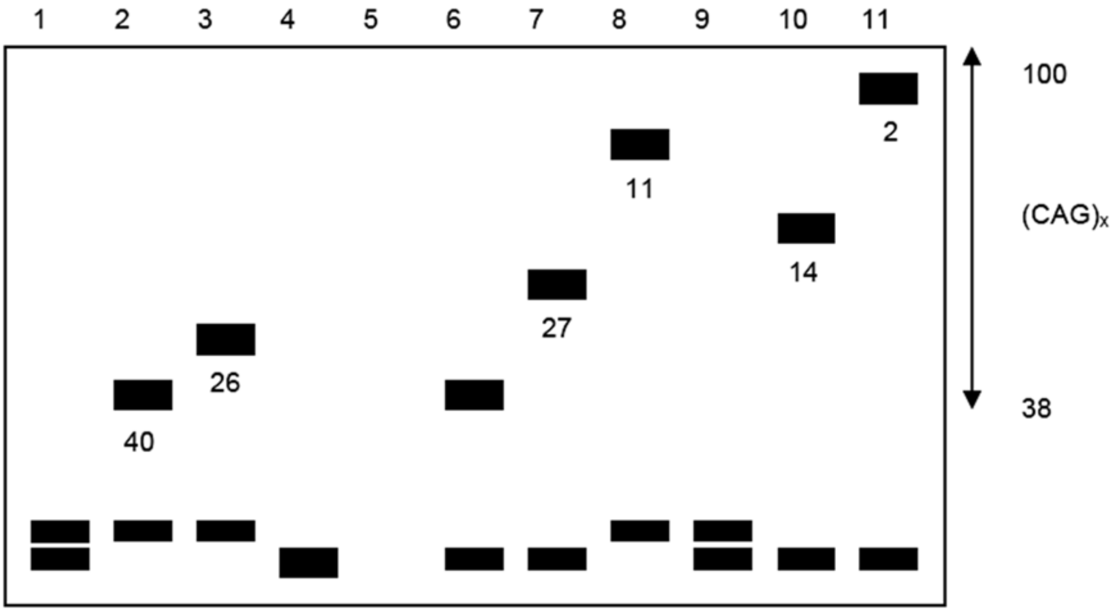
C Analyse each marker by PCR rather than RFLP analysis.

D Select markers present on the sex chromosomes rather than on the autosomes.

37 The data below shows the results of electrophoresis of PCR fragments amplified using primers for the site that has been shown to be altered in Huntington's disease.

The inherited mutation in the Huntington's disease gene abnormally repeats the nucleotide sequence CAG from 36 up to more than 120 times of that. The male parent, shown as individual 2, had the onset of Huntington's disease when he was 40 years old.

Six of his children (individuals 3, 5, 7, 8, 10, 11) suffer from Huntington's disease, and the age at which the symptoms first began is shown by the number below the band from the PCR fragment.



What is the likely outcome for the normal individuals 4, 6, and 9?

A Individuals 4 and 9 do not have the trait, and will not get Huntington's disease, but individual 6 is likely to start the disease when he reaches his father's age of 40.

B Individuals 4, 6, and 9 have not inherited the defect causing Huntington's disease.

C Individuals 4, 6, and 9 will still develop Huntington's disease at some point in their lives, since the disease is inherited as a dominant trait.

D Two of the three will develop the disease, since it is inherited as a dominant trait, but the data does not allow you to predict which two.

38 Which of the following statements are true about adult stem cells?

1 They can undergo self-renewal

2 They can be totipotent, pluripotent or multipotent.

3 They can differentiate into almost any cell type.

4 They can give rise to specialised cells.

A 1 and 2 **B** 1 and 4 **C** 2 and 3 **D** 3 and 4

39 Which of the following methods can be used to introduce a foreign gene into a plant cell?

1 By Agrobacterium-mediated transfection

2 By microinjection of naked DNA

3 Using a phage delivery vector

4 Using a gene gun

A 1 and 3 only

B 1, 2 and 4 only

C 1, 3 and 4 only

D All of the above

40 Developing fish eggs can be treated to produce a diploid egg. In salmon, such eggs have been fused with haploid salmon sperm to give infertile triploid salmon.

Reproductive organ tissue from diploid trout was transplanted into newly hatched triploid salmon. This tissue matured as the fish grew and the salmon successfully produced viable trout sperm or eggs which resulted in young trout.

Which fish could be seen as genetically modified?

A salmon providing eggs for treatment

B trout providing reproductive organ tissue

C young triploid salmon

D young trout